



Syllabus

Fall Semester 2027 (샘플)

Course Title	Reinforcement Learning	Course Code	MC0000	Credit	3	Classroom	5701
		Type	Lecture	Hours	3		

Teacher	Name	Office	Office hour	Contact	E-mail
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1. Course Descriptions

- To formalize a decision-making problem as a Markov Decision Process (MDP) and derive the Bellman equations
- To implement and compare tabular reinforcement learning algorithms: dynamic programming, Monte Carlo methods, and temporal-difference learning
- To understand and implement deep reinforcement learning methods that combine function approximation with neural networks (DQN, policy gradient methods, PPO)
- To understand the principles behind modern applications such as imitation learning and RLHF (Reinforcement Learning from Human Feedback)
- To apply reinforcement learning algorithms to a simulated environment as a team, following an honest experimental protocol (multiple seeds, baseline comparison)

This course is a follow-up to "Machine Learning & Deep Learning" and assumes prior familiarity with neural networks.

2. Core Target Competencies

Subject Core Competencies			
Fundamental Reinforcement Learning and Problem-Solving Skills	Logical Thinking and Analytical Skills	Communication and Collaboration Skills	Ethical Use of AI and Responsible Attitude
• Demonstrates understanding of core reinforcement learning algorithms and applies them to a simulated decision-making problem. • Derives the mathematical principles behind algorithms (e.g. Bellman equations, policy gradients) and uses them to explain why a result occurred. • Clearly explains a team project's design and results to others, and gives constructive peer-review feedback to other teams. • Reports experimental results honestly (including unstable seeds or failed attempts), and uses AI tools responsibly in coursework.			

3. Textbooks and References

A. Textbook: None (slides and handouts)

B. References

- "Reinforcement Learning: An Introduction" (2nd ed.), Richard S. Sutton and Andrew G. Barto
- ★ There is no main textbook. Main materials can be downloaded from the Google Classroom.

4. Class Method

A. Google Classroom code : [TBD]

B. Lecture slides and code examples are distributed through the Google Classroom.

5. Evaluation

A. Evaluation Criteria

Quizzes & Homework	Midterm Exam.	Final Exam.	Team Project	Attendance and Attitudes	Sum
20	25	25	25	5	100

B. Evaluation Method

1. Absolute evaluation
2. Weekly quiz and homework for each class
3. Midterm exam: written test (2 hrs)
4. Final exam: written test (2 hrs)
5. Team project: two capstone projects (Week 8: tabular RL, Week 16: deep RL), each graded on a written report, a presentation, and peer-review. Grading emphasizes the honesty of the experimental protocol (seeds ≥ 3 , baseline comparison against a random policy, honest reporting of unstable results) over raw performance.

6. Criteria for Awarding Credits

- **Pass:** In the case of receiving a score of **70 or more out of 100** according to the criteria of the evaluation method.
- **Academic Excellence Award:** Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.
- **Attendance:** At least **three-quarters** of the total class hours is required to earn credit.
- **Pass/Fail courses:** If the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.
- **Excused absences:** For official duties or medical reasons, the make-up score for a missed exam shall be calculated as follows: the percentile score corresponding to the student's total score in other assessments of the same course shall be applied to the missed exam, multiplied by a weighting factor (**100%** for official absences, **80%** for medical absences). If there are no available reference scores, a re-assessment shall be conducted.

※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

7. Weekly Plans

(Exact dates will be announced once the 2027 academic calendar is confirmed.)

Weeks	Date	Topic and Contents	Note
Week 1	TBD	Introduction to RL, Neural Network Review	
Week 2	TBD	Multi-Armed Bandits	
Week 3	TBD	MDP Formalization	
Week 4	TBD	Dynamic Programming	
Week 5	TBD	Monte Carlo Methods	
Week 6	TBD	Temporal-Difference Learning	
Week 7	TBD	n -step Bootstrapping, Eligibility Traces, Planning	
Week 8	TBD	Block A Capstone: Team Project	Midterm exam
Week 9	TBD	Function Approximation and DQN	
Week 10	TBD	Policy-Based Reinforcement Learning	
Week 11	TBD	Advanced Policy Optimization: PPO	
Week 12	TBD	Imitation Learning and RLHF	
Week 13	TBD	Robot Simulation and Control Basics	
Week 14	TBD	Advanced Simulation: MuJoCo, Isaac Sim	
Week 15	TBD	Model-Based RL and Monte Carlo Tree Search	
Week 16	TBD	Block B Capstone: Team Project	Final exam

※ The above plans may change according to the academic schedule.